

Dynamic Sensor Orientation Unit for the Intelligent Mobility Cane

Manu Suresh

Golisano Computing and
Information Sciences
Rochester Institute of
Technology
Rochester, New York, USA
ms5529@rit.edu

Jagannadh Pariti

Golisano Computing and
Information Sciences
Rochester Institute of
Technology
Rochester, New York, USA
jsp3721@g.rit.edu

Tae Oh

Golisano Computing and
Information Sciences
Rochester Institute of
Technology
Rochester, NY, USA
thoics@rit.edu

ABSTRACT

Numerous studies have been conducted to improve the features of the “smart” canes. However, there is insufficient research performed to develop the guidelines to create a usable smart cane. In this paper, a Dynamic Sensor Orientation Unit (DSOU) is introduced to maintain a detection distance irrespective of the angle of the cane. The paper also aims to outline guidelines for placement of the Obstacle Detection System (ODS) and the sensor angle to be used in our Intelligent Mobility Cane (IMC) prototypes. A usability test with 11 people who use a white cane for daily navigation was conducted to study the users’ preference and behavior with a fixed sensor angle IMC. The analysis revealed the angle of the sensor is dependent on the participant’s height and how users hold the canes. The results suggest requirements for sensor angle orientation and outline guidelines for ODS designs on future IMCs.

Author Keywords

Accessibility; Intelligent Mobility Cane; Ultrasonic Sensors; Sensor Angle; Dynamic Orientation; SmartCane; Visually Impaired;

ACM Classification Keywords

• **Human-centered computing~Haptic devices** • Human-centered computing~User studies • **Social and professional topics~People with disabilities**

INTRODUCTION

A white cane is one of the most popular assistive tools due to its low cost and convenience. However, the cane has several limitations and issues. For example, any obstacles that are above knee-level and sudden drop-offs are difficult

to detect using a regular white cane. Further, the white cane detects obstacles at a distance equal to the cane’s length [3]. Today, smart canes utilize ultrasonic sensors to detect obstacles that are above knee-level, extend the obstacle detection range of the cane and automatically guide users away from obstacles in their path. The users are informed through auditory or tactile feedback from the smart canes [2,3,4,5,8,9,10]. However, these smart canes force the user to change their regular grasping technique. Balakrishnan et al. investigated the different grips used by cane users to hold their canes [1]. According to the National Association for the Blind (NAB), the angle that the cane makes with the ground varies between 50° and 60° [6]. This problem was addressed by them by allowing the users to manually adjust the sensor angle between two set positions. This was calculated based on the average angle of 55° [1]. However, issues such as usability, affordability, and availability have made it difficult for people who are visually impaired to adopt smart canes. Issues identified by Kim et al. [6] were, 1) Battery Issues, 2) Reaction Time and 3) Floor-level detection. This paper focuses on addressing floor-level detection issues. The paper introduces a Dynamic Sensor Orientation Unit (DSOU) that maintains a fixed detection distance from the cane irrespective of the angle the cane makes with the floor.

DYNAMIC SENSOR ORIENTATION UNIT

The Intelligent Mobility Cane (IMC) [10] is a 54-inch regular foldable white cane with a forward sensor mounted towards the tip of the cane rather than the handle. The IMC was designed to be flexible with sensor angle and haptic notification devices as well as to be inexpensive and light weight. The Obstacle Detection System (ODS) consists of an Arduino UNO microcontroller for processing sensor data from a Maxbotix MB1240 ultrasonic sensor placed 1 foot from the tip of the cane. Two vibration mini-disk motors mounted on the handle provide haptic feedback to the user. The ultrasonic sensor was set to detect objects that were 4 feet from the sensor, effectively making the detection distance 3 feet [7,11]. The Dynamic Sensor Orientation Unit (DSOU) consists of a Particle Photon microcontroller which processes the angle data from the (FLORA 9-DOF) accelerometer sensor and controls the (HXT900) servo

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Motor. The ultrasonic sensor was mounted on the servo motor arm. Figure 1 shows the mounted ODS and DSOU on a standard white cane. The formula used to calculate the sensor angle Θ° was dependent on the angle at which the user held the cane. The formula is derived from figure 2.

$$\Theta^\circ = \sin^{-1}((0.25) \sin(\alpha)) * (57296 / 1000)$$

The photon microcontroller will send the Θ° to the servo to adjust the angle of the sensor.

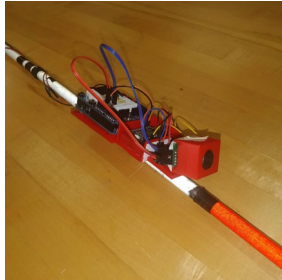


Figure 1: ODS and DSOU mounted on a collapsible cane

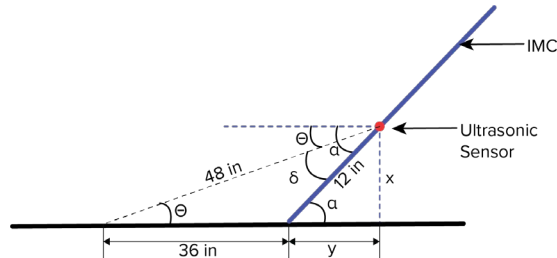


Figure 2: The angle of the sensor (δ) is calculated using the cane angle (α) and the angle the ultrasonic beam makes with the ground (Θ) at 48 inches from the ultrasonic sensor.

METHOD

Eleven participants (7 men, 4 women) were recruited with the help of the Association for the Blind and Visually Impaired (ABVI) in Rochester, NY. When each participant arrived, a description of the study, IMC and obstacle path were provided. They were then asked a set of pre-task interview questions. The task was to navigate an obstacle path set up in a conference room at ABVI's Rochester, NY office to get feedback from the participant about feedback on the fixed sensor IMC. The DSOU was set to constantly send angle data to a team members laptop. The obstacle path consisted of a long table, resting on its side, in the middle of the room with three small plastic bins, approximately 1.5 ft high and 1ft wide, as obstacles. Two of the bins were placed to the left of the table, and the other was placed to the right of the table. Audio and video were captured while the participants navigated the obstacle path. After completing the obstacle path, each participant was asked post-task interview questions.

RESULTS AND DISCUSSION

All participants were able to navigate the path with minimal contact with the obstacles. The angle data collected indicated in Table 1, shows that the cane for each participant varied notably throughout the task.

Effect of Height, Cane Length and Grip on Cane Angle

Any changes to the angle of the cane directly affect the ability of the ultrasonic sensor to detect obstacles. The factor that affects the angle of the cane the most is the cane grasping technique. The cane angle is not only affected by the different grasp, but also by the same techniques employed at different points on the cane handle. All the participants mentioned changing the grasping technique or changing the position along with the cane handle based on the familiarity of the environment, weather condition, and terrain. Another factor that affects the angle of the cane is cane length. Participant 4 uses a cane that is longer than is recommended for more reach. This makes the angle the cane makes with the ground more acute. This effectively reduces the detection distance of the ODS, giving the user less time to react to any obstacle in their path. The need for the DSOU is shown in Table 1 by the variation in angle for each participant. The DSOU ensures that the detection distance is maintained irrespective of the grasping technique, cane length and angle of the cane.

Users	Mean $^\circ$	σ°	Height	Grasp
P1	54.1	14.76	160.02	Pencil
P2	58.28	16.62	182.88	Handshake/Pencil
P3	61.73	11.36	162.56	Handshake/Vertical
P4	67.71	9.74	167.64	Handshake
P6	56.09	10.51	157.48	Handshake/Vertical
P7	64.34	11.27	177.8	Handshake
P8	62.44	10.14	180.34	Handshake/Pencil
P9	51.85	7.53	175.26	Pencil
P10	65.85	12.79	162.56	Handshake/Pencil
P11	54.2	5.62	190.5	Pencil

Table 1: Mean and standard deviation of cane angle for each participant

Design Considerations

Variable sensor angles. Different users prefer different cane angles (with the floor). Further, a single user may prefer different cane angles under different circumstances. Therefore, no one “standard” angle should be considered or pursued in the design of IMCs.

Detection Distance. Most of the participants said that the detection distance seemed right. Participant 2 felt that the detection distance was set far enough for them to avoid the obstacle. However, Participant 4 wanted the distance set as far as possible to facilitate seamless avoidance of obstacles. Participants 6 asked if the detection distance can be set by the user. Hence, the detection distance must be variable.

Collapsible canes. All the participants preferred the ability to collapse the canes they use. Therefore, the design of a good IMC should try to preserve this capability.

CONCLUSION AND FUTURE WORK

This study aims to provide insight into the issues faced while using a fixed sensor angle IMC and introduces the DSOU to alleviate the issues. A future dynamic versus fixed sensor study could reveal possible improvements in performance.

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